

System Architecture Specification: AI Study Companion

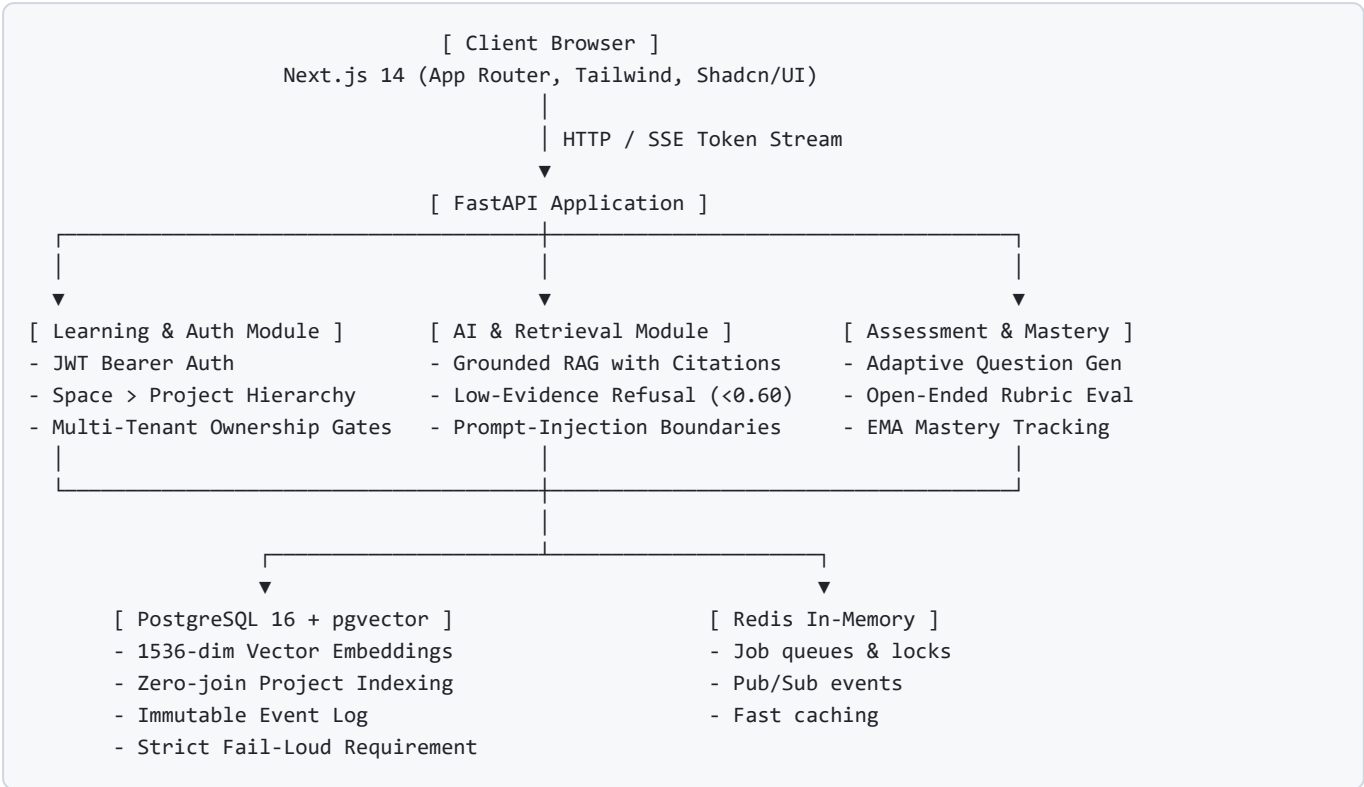
System Name: AI Study Companion

Architecture Pattern: Clean Modular Hexagonal Architecture with Strict Tenant Isolation

Runtime: Next.js 14 (Frontend) + FastAPI ASGI (Backend) + PostgreSQL 16 with pgvector + Redis

1. High-Level Architecture Overview

The system is architected across decoupled layers ensuring zero leakage between learning projects, real-time citation tracking, and continuous mastery estimation:



2. Core Relational & Vector Schema

All project-scoped entities enforce a direct `project_id` foreign key index to guarantee query isolation without complex joins.

Key Domain Models

1. **User** : `id` , `email` , `hashed_password` (bcrypt), `name` , `role` (`user` , `admin`).

2. **Space** : Top-level learning namespace (`user_id` , `name` , `description`).
3. **Project** : Core study boundary (`space_id` , `user_id` , `name` , `learning_goal`).
4. **Material** : Uploaded learning resource (`project_id` , `filename` , `file_path` , `status` , `page_count`).
5. **DocumentChunk** : Semantic text segment with 1536-dimensional vector embedding (`material_id` , `project_id` , `page_number` , `content` , `embedding Vector(1536)`).
6. **Concept** : Knowledge atom extracted from materials (`project_id` , `name` , `description`).
7. **ConceptMastery** : Real-time mastery state (`project_id` , `user_id` , `concept_id` , `mastery_score` [0.0-1.0] , `confidence_level` , `trend` [IMPROVING|STABLE|ATTENTION]).
8. **MasterySnapshot** : Historical timestamped mastery point for progress graphing.
9. **Conversation** & **Message** : Dialogue history with JSON citation provenance tags (`filename` , `page_number` , `similarity`).
10. **QuizSession** , **Question** , **Answer** : Diagnostic assessments with rubric evaluations.
11. **LearningContext** : Persistent learner summary (strengths, weak concepts, repeated mistakes).
12. **LearningEvent** : Immutable event stream with unique `idempotency_key` .
13. **AIRequestLog** : Telemetry capturing prompt tokens, completion tokens, latency, cost USD, and status (`SUCCESS` , `REFUSED_LOW_EVIDENCE`).
14. **Recommendation** : Actionable next step answering "What should I do next?".

3. Grounded Retrieval & Guardrail Mechanics

1. **Embedding Generation**: User query embedded to 1536 dimensions.
2. **Project-Scoped Vector Search**: $\text{similarity} = 1.0 - \text{cosine_distance}(\mathbf{v}_{\text{query}}, \mathbf{v}_{\text{chunk}})$ filtered strictly by `project_id` .
3. **Low-Evidence Refusal Guardrail**:
 - If $\max(\text{similarity}) < 0.60$ or no chunks match: Refusal triggered: "I do not have sufficient information in your uploaded project materials to answer this question reliably."
 - Logged with `status="REFUSED_LOW_EVIDENCE"` .
4. **Provenance Citations**:
 - Injected into system prompt context with tags: `[Source: {filename} – Page {page_number}]` .
 - Rendered in frontend as interactive badge chips.

4. Adaptive Assessment & EMA Mastery Updates

- **Target Concept Selection**: Prioritizes concepts tagged as `ATTENTION` or score < 0.50 (60% weight).
- **Exponential Moving Average (EMA)** update formula: $\text{Mastery}_{\text{new}} = 0.3 \cdot \text{Score}_{\text{new}} + 0.7 \cdot \text{Mastery}_{\text{prev}}$
- **Trend Classification**:
 - $\Delta \geq +0.10 \implies \text{IMPROVING}$
 - $\text{Score} < 0.50$ or $\Delta \leq -0.10 \implies \text{ATTENTION}$
 - Otherwise $\implies \text{STABLE}$